

chap_simples_comparaison

2025-07-09

Étude de la stratégie (SI, HT) par simulations de Monte-Carlo

```
require("tidyverse")###pour ggplot
```

```
## Loading required package: tidyverse
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr    1.5.1
```

```
## v ggplot2     3.5.2      v tibble     3.3.0
```

```
## v lubridate  1.9.4      v tidyr      1.3.1
```

```
## v purrr       1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
require(gridExtra)##pour combiner des graphiques
```

```
## Loading required package: gridExtra
```

```
##
```

```
## Attaching package: 'gridExtra'
```

```
##
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
##      combine
```

```
library(sampling)
```

```
data(rec99)
```

```
N <- nrow(rec99)
```

```
n <- 70
```

```
n.sim <- 10000
```

```
est.si <- numeric(n.sim)
```

```
piks <- rep(n/N,n)
```

```
set.seed(123456)
```

```
for (i in 1:n.sim)
```

```
{
```

```
  si.rec <- srswor(n,N)
```

```
  ys <- rec99$LOGVAC[si.rec==1]
```

```
  est.si[i] <- HTestimator(ys,piks)
```

```
}
```

```
mean(est.si)
```

```
## [1] 10799.5
```

```
# [1] 10757.02
(cv.si <- sd(est.si)/mean(est.si))
```

```
## [1] 0.1896896
```

```
# [1] 0.1918109
```

```
ty <- sum(rec99$LOGVAC)
(mean(est.si)-ty)/ty
```

```
## [1] 0.002925289
```

```
# [1]-0.001020021
```

histogramme des 10000 simulations

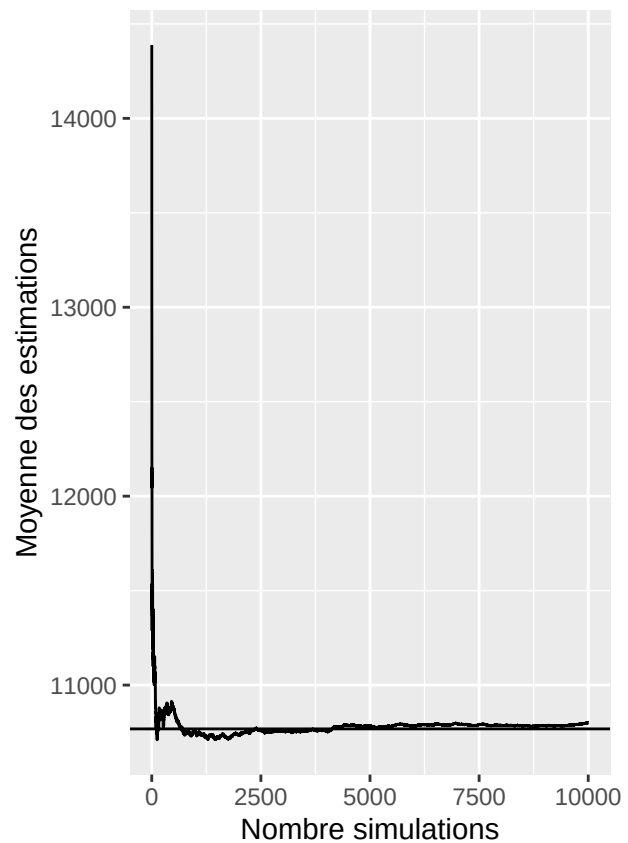
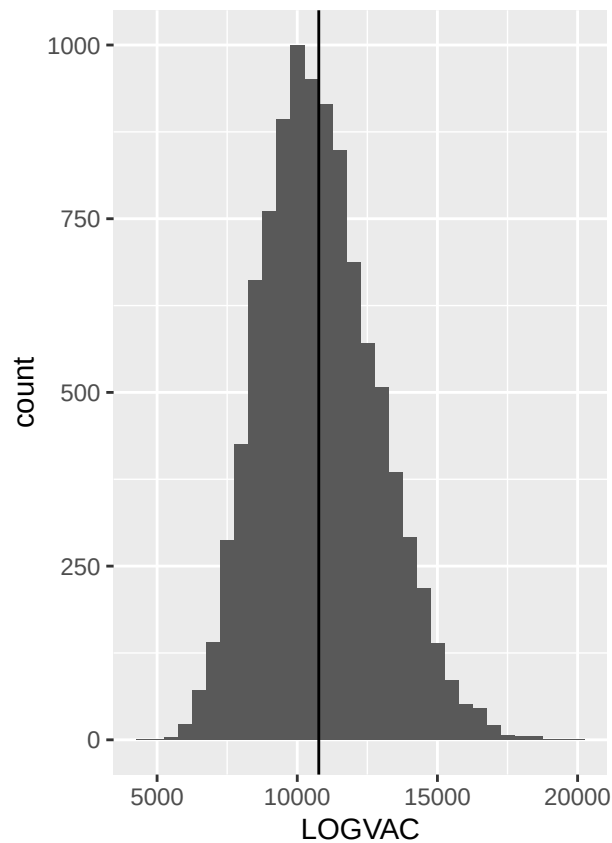
```
est.si <- data.frame(est.si)
colnames(est.si)=c("LOGVAC")

p1 <- ggplot(est.si,aes(x=LOGVAC))+
  geom_histogram(binwidth=500)+
  geom_vline(xintercept=10768)
  #ggtitle("10000 estimations par l'estimateur HT, plan SI")

#####biais de l'estimateur HT
nb.simul<-matrix(1:10000,10000,1)
est.si.cumul <- cumsum(est.si$LOGVAC)/nb.simul
df.si <- data.frame(nb.simul, est.si.cumul)
#View(df.si)

ytot <- sum(rec99$LOGVAC)
p2 <- ggplot(df.si,aes(x=nb.simul,y=est.si.cumul))+
  geom_line()+
  geom_hline(yintercept=ytot)+
  xlab("Nombre simulations")+
  ylab("Moyenne des estimations")

###mettre les deux graphiques ensemble
grid.arrange(p1,p2,nrow=1)
```



Étude de la stratégie (BE, HT) par simulations de Monte-Carlo

```
BE1 <- function(N,Pi)
{
  u <- runif(N, 0, 1)
  as.numeric(u<Pi)
}

est.be <- numeric(n.sim)
n.be <- numeric(n.sim)
set.seed(123456)
for (i in 1:n.sim)
{
  s.be <- BE1(N,n/N)
  n.be[i] <- sum(s.be==1)
  rec.be <- rec99[s.be==1,]
  est.be[i] <- (N/n)*sum(rec.be$LOGVAC)
}
mean(est.be)

## [1] 10793.04

# [1] 10764.34
(cv.be <- sd(est.be)/mean(est.be))

## [1] 0.2189063
```

```
# [1] 0.2193195
(mean(est.be)-ty)/ty

## [1] 0.002325544
# [1]-0.000339896

quantile(n.be,probs=c(0.025,0.1,0.5,0.9,0.95,0.975))

## 2.5% 10% 50% 90% 95% 97.5%
## 55 60 70 80 83 85
# 2.5% 50% 55 60 90% 95% 97.5% 70 80 83 85
```

Étude de la stratégie (SY, HT) par simulations de Monte-Carlo

```
pik <- rep(n/N,N)
est.sy <- numeric(n.sim)
set.seed(123456)
for (i in 1:n.sim)
{
  sy.rec <- UPsystematic(pik)
  ys <- rec99$LOGVAC[sy.rec==1]
  piks <- pik[sy.rec==1]
  est.sy[i] <- HTestimator(ys,piks)
}
mean(est.sy)

## [1] 10766.32
# [1] 10774.91
(cv.sy <- sd(est.sy)/mean(est.sy))

## [1] 0.34274
# [1] 0.1812545
(mean(est.sy)-ty)/ty

## [1] -0.0001561964
# [1] 0.0006417162

orderlog <- order(rec99$LOG)
rec.tri <- rec99[orderlog,]
```